

Project One

Missing values per column:

```
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
CouponCode       309
ReferralSource   0
TotalPrice       0
dtype: int64
```

Missing values after fix:

```
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
CouponCode       0
ReferralSource   0
TotalPrice       0
dtype: int64
```

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	OrderID	1200 non-null	object
1	Date	1200 non-null	object
2	CustomerID	1200 non-null	object
3	Product	1200 non-null	object
4	Quantity	1200 non-null	int64
5	UnitPrice	1200 non-null	float64
6	ShippingAddress	1200 non-null	object
7	PaymentMethod	1200 non-null	object
8	OrderStatus	1200 non-null	object
9	TrackingNumber	1200 non-null	object
10	ItemsInCart	1200 non-null	int64
11	CouponCode	1200 non-null	object
12	ReferralSource	1200 non-null	object
13	TotalPrice	1200 non-null	float64

dtypes: float64(2), int64(2), object(10)

memory usage: 131.4+ KB

None

Project Two

First 5 rows:

	OrderID	Date	CustomerID	...	CouponCode	ReferralSource	TotalPrice
0	ORD200000	2023-01-04	C72649	...	SAVE10	Instagram	2853.10
1	ORD200001	2024-08-23	C75739	...	SAVE10	Referral	302.70
2	ORD200002	2024-02-27	C81728	...	FREESHIP	Email	2753.40
3	ORD200003	2023-10-15	C33540	...	SAVE10	Facebook	273.19
4	ORD200004	2025-05-08	C81840	...	SAVE10	Email	2504.04

[5 rows x 14 columns]

Missing values:

OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0
dtype:	int64

BASIC STATISTICS

```
=====
              count      mean  median   min     max
Quantity    1200.0      2.95    3.00    1.00    5.00
UnitPrice   1200.0    356.41   364.21   11.39   699.93
TotalPrice  1200.0   1053.97   823.62   11.39  3456.40
=====
```

OUTLIERS (IQR METHOD)

```
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Lower bound: -1341.41
Upper bound: 3330.41
Number of outliers: 8
Sample outliers:
```

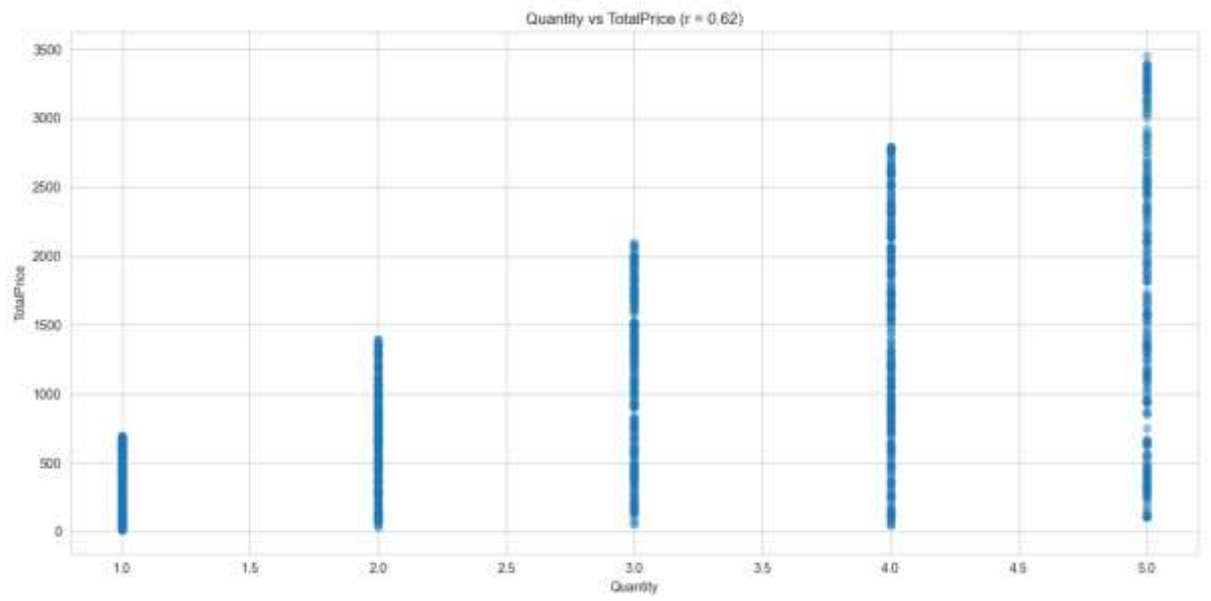
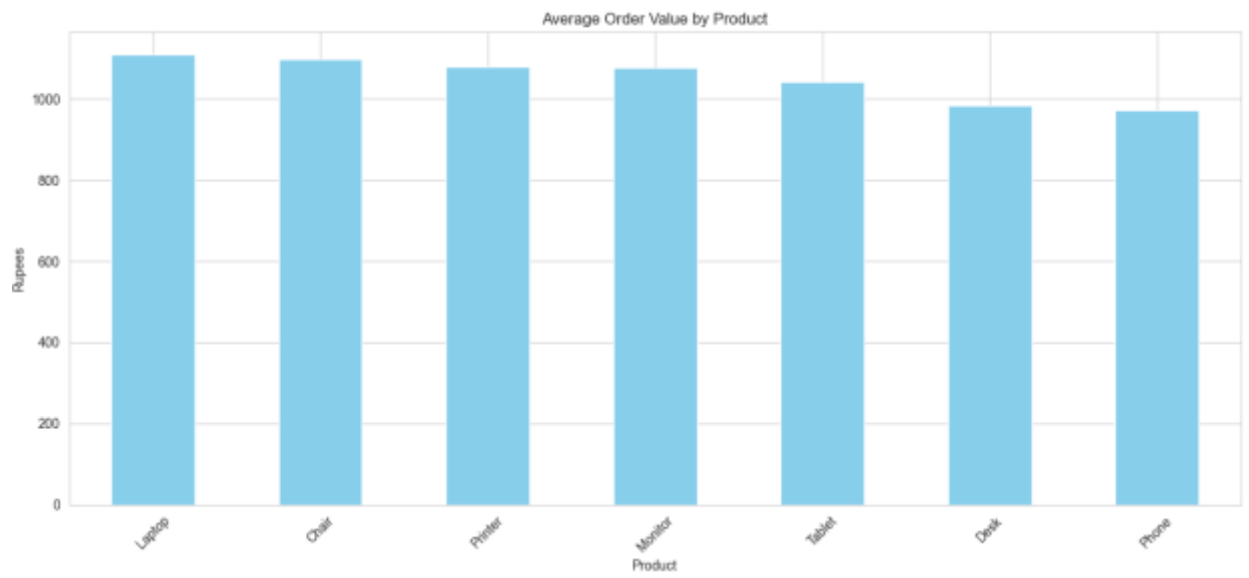
	OrderID	Product	Quantity	TotalPrice
107	ORD200107	Printer	5	3353.75
326	ORD200326	Laptop	5	3352.40
328	ORD200328	Tablet	5	3370.20
469	ORD200469	Chair	5	3384.90
632	ORD200632	Laptop	5	3390.80

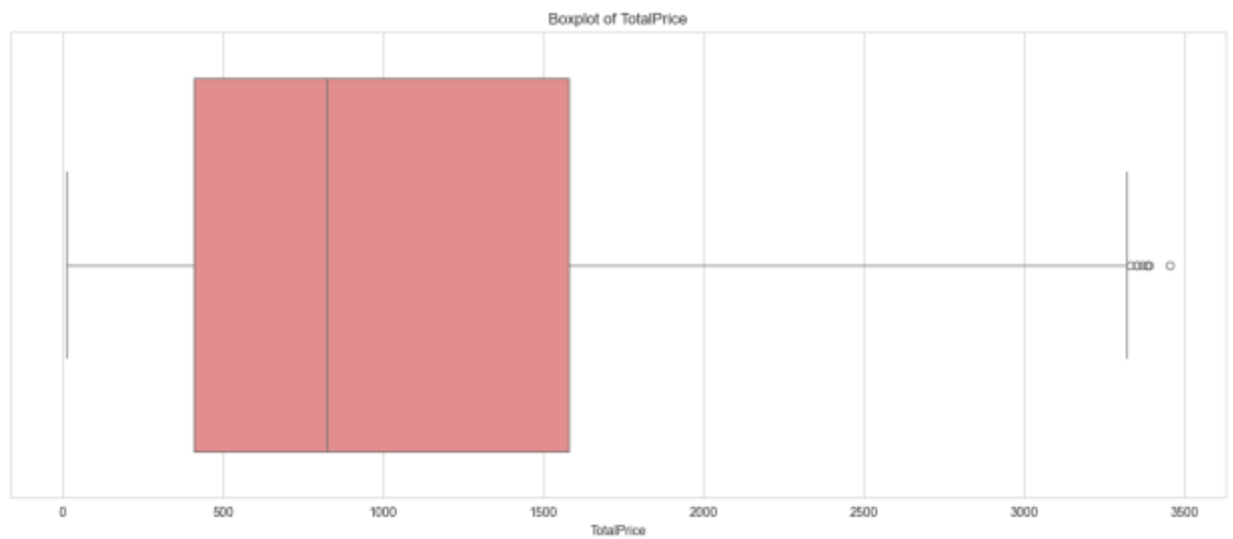
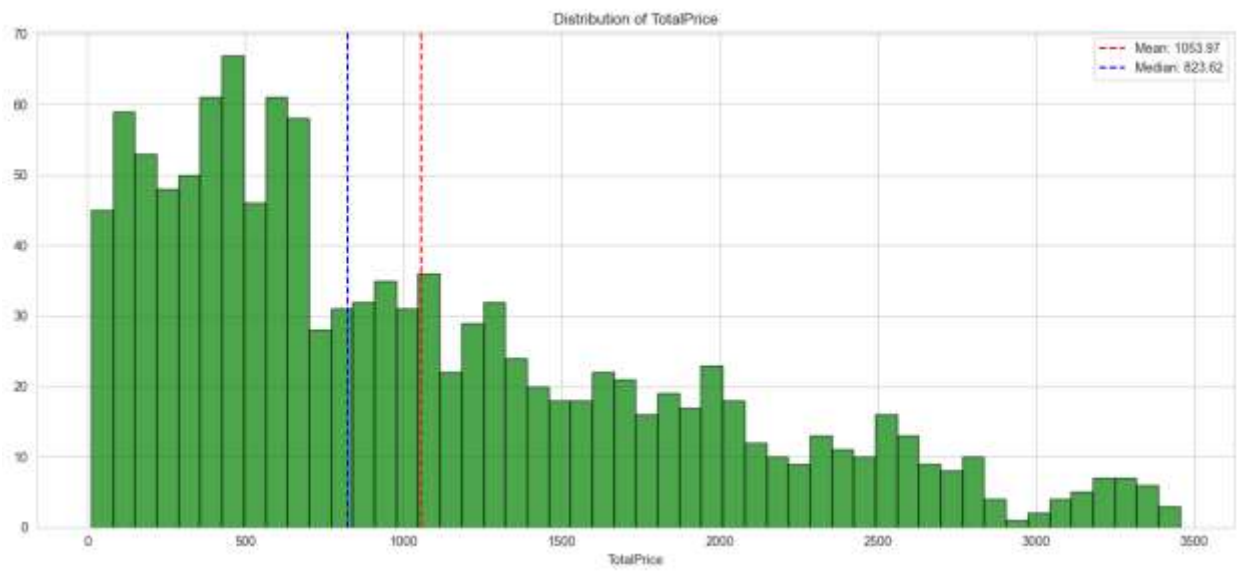
AVERAGE ORDER VALUE BY PRODUCT

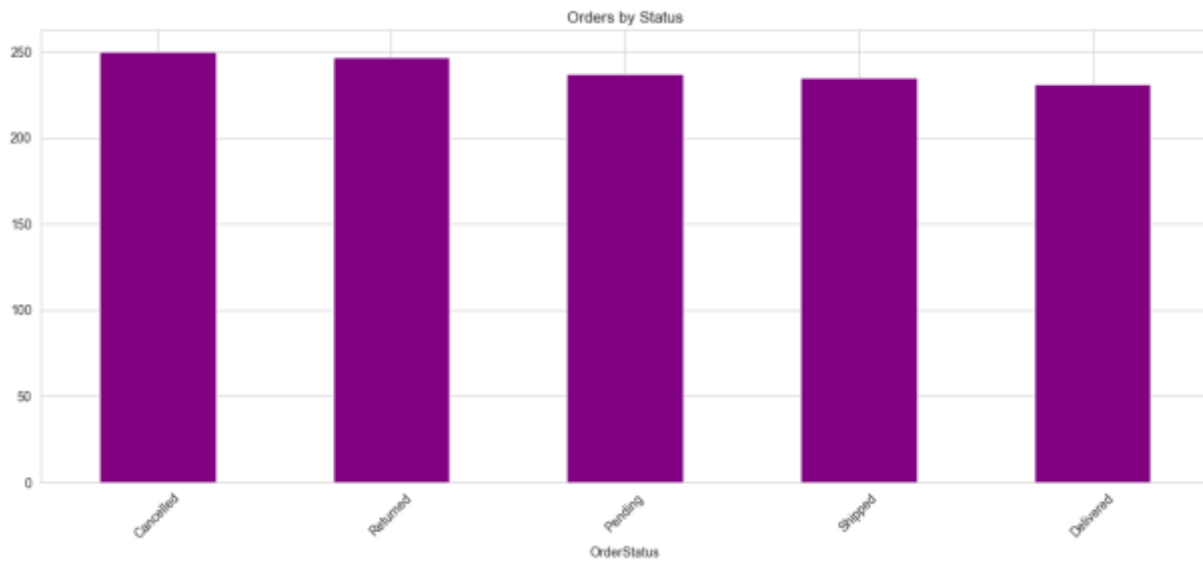
Product

Laptop	1110.56
Chair	1098.99
Printer	1080.73
Monitor	1077.62
Tablet	1042.28
Desk	985.06
Phone	972.58

Name: TotalPrice, dtype: float64







Project Three

Dataset shape: 1200 rows, 14 columns

Date range: 2023-01-01 to 2025-06-30

SQLite in-memory database created with table 'orders'

1. OVERALL STATISTICS

total_orders	total_revenue	avg_order_value	min_order_value	max_order_value
1200	1264761.96	1053.97	11.39	3456.4

2. REVENUE BY PRODUCT

Product	units_sold	revenue	avg_unit_price	order_count
Chair	562	195620.11	355.66	178
Printer	542	195612.61	351.71	181
Laptop	535	192126.56	357.71	173
Tablet	497	186568.95	367.68	179
Monitor	480	175651.41	358.66	163
Desk	508	167459.93	329.61	170
Phone	411	151722.39	375.22	156

3. ORDER STATUS DISTRIBUTION

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OrderStatus	num_orders	percentage
Cancelled	250	20.83
Returned	247	20.58
Pending	237	19.75
Shipped	235	19.58
Delivered	231	19.25

=====

4. HIGH-VALUE DELIVERED ORDERS (WHERE TotalPrice > 1000)

=====

OrderID	Date	Product	Quantity	TotalPrice	PaymentMethod
ORD200789	2023-08-17	Tablet	5	3456.40	Online
ORD200632	2023-05-02	Laptop	5	3390.80	Gift Card
ORD201065	2023-10-30	Printer	5	3334.00	Debit Card
ORD200361	2024-06-29	Printer	5	3299.25	Credit Card
ORD200837	2025-06-14	Chair	5	3277.75	Online
ORD201079	2024-06-20	Phone	5	3170.00	Gift Card
ORD200511	2024-03-15	Monitor	5	2876.20	Cash
ORD200578	2024-10-10	Monitor	5	2830.35	Credit Card
ORD200883	2025-05-13	Printer	5	2807.40	Online
ORD200769	2023-07-06	Printer	4	2714.20	Cash

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5. MONTHLY SALES TREND (2023-2025)

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month	num_orders	revenue	avg_order_value
2023-01	47	56685.75	1206.08
2023-02	37	40117.66	1084.26
2023-03	43	48609.37	1130.45
2023-04	31	27751.71	895.22
2023-05	49	63836.84	1302.79
2023-06	45	49500.19	1100.00
2023-07	44	42820.66	973.20
2023-08	51	54352.14	1065.73
2023-09	29	29526.67	1018.16
2023-10	47	52607.85	1119.32
2023-11	41	43079.67	1050.72
2023-12	46	43754.73	951.19
2024-01	32	38528.08	1204.00
2024-02	32	36909.57	1153.42
2024-03	36	36030.90	1000.86
2024-04	50	49613.14	992.26
2024-05	34	27909.11	820.86
2024-06	53	68068.54	1284.31
2024-07	43	42963.98	999.16

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2024-08	28	31991.07	1142.54
2024-09	44	39794.98	904.43
2024-10	31	37226.97	1200.87
2024-11	35	32413.76	926.11
2024-12	41	38785.77	945.99
2025-01	27	29099.40	1077.76
2025-02	37	35317.55	954.53
2025-03	49	39200.66	800.01
2025-04	32	31821.20	994.41
2025-05	37	43396.64	1172.88
2025-06	49	53047.40	1082.60

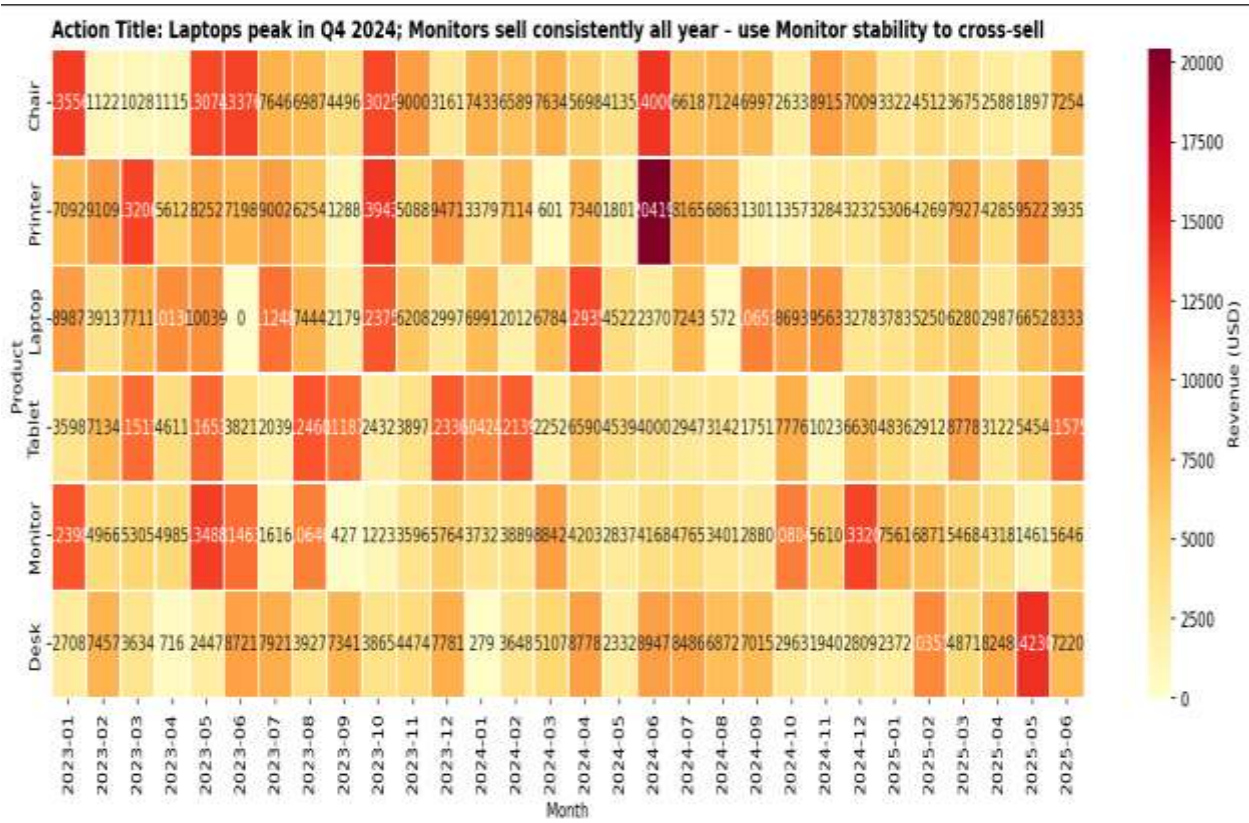
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6. PAYMENT METHOD BREAKDOWN

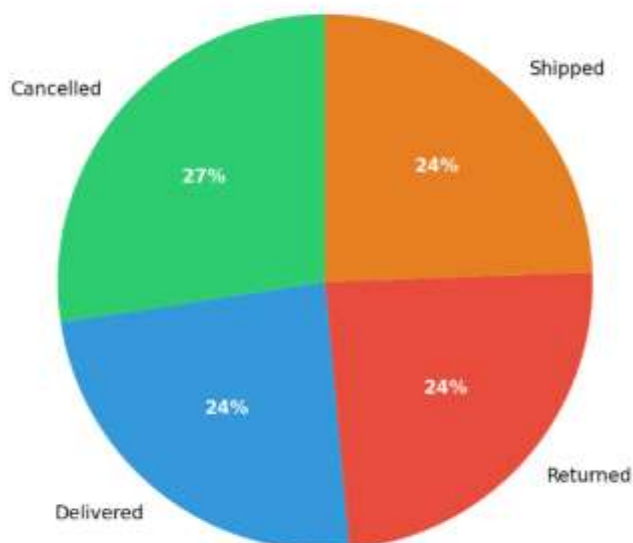
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PaymentMethod	usage_count	total_revenue	avg_order_value
Credit Card	234	263847.63	1127.55
Online	258	262442.94	1017.22
Cash	246	259786.29	1056.04
Gift Card	230	246323.92	1070.97
Debit Card	232	232361.18	1001.56

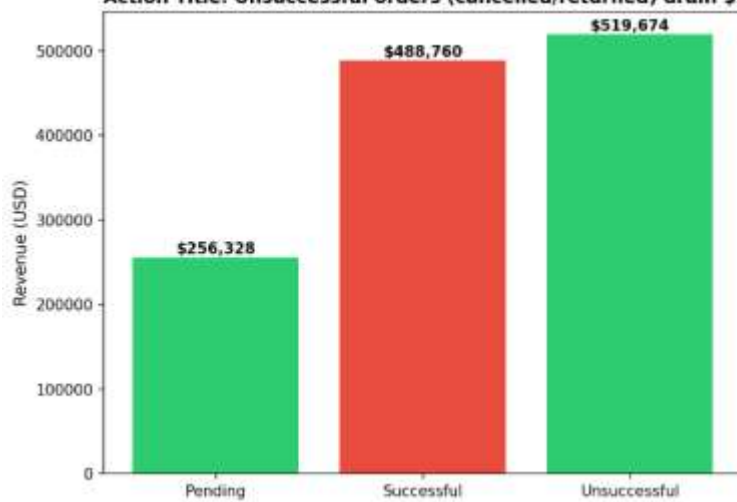
Project Four

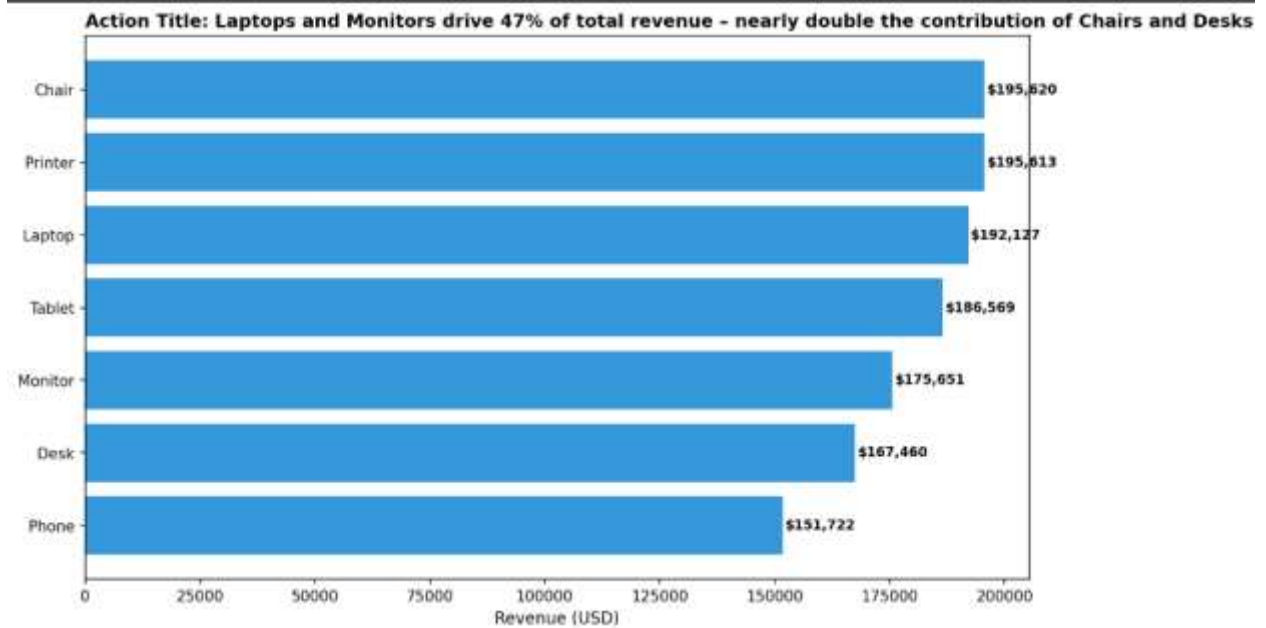
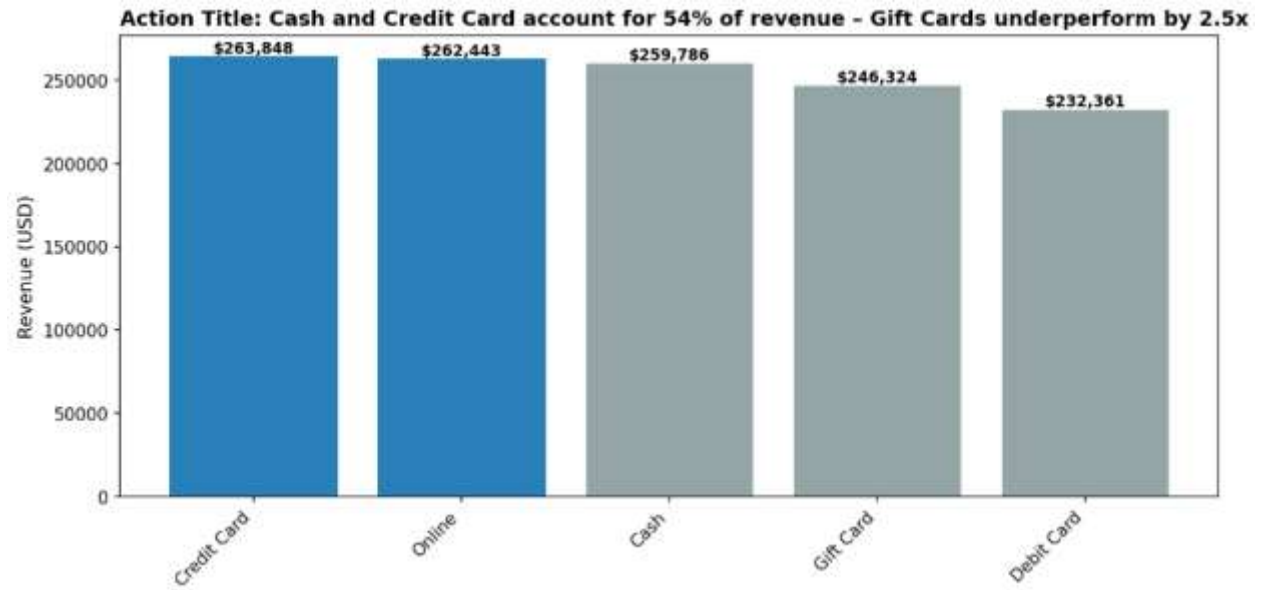


Action Title: Nearly 30% of revenue is lost to Cancelled (17%) or Returned (12%) orders - a \$140K leakage



Action Title: Unsuccessful orders (cancelled/returned) drain \$140,000 - fix returns to unlock immediate growth





Action Title: Instagram, Facebook, and direct Referrals generate >70% of referral revenue - Email and Google lag

